



GOA COLLEGE OF ENGINEERING

"BHAUSAHEB BANDODKAR TECHNICAL EDUCATION COMPLEX"
FARMAGUDI, PONDA- GOA - INDIA

MATHEMATICS-III

UNIT 1: MATRICES

LECTURE 9

DEVELOPED BY MATHEMATICS FACULTY
DEPARTMENT OF SCIENCE & HUMANITIES
GOA COLLEGE OF ENGINEERING



MATRICES LECTURE 9

TOPICS TO BE COVERED:

- Functions of a Matrix
- Minimal polynomial of a matrix



Functions of a matrix

Applications of Diagonalisation of matrix:

Given a square matrix **A**, to find the functions $A^n, e^A, \sin A$ and $\cos A$:

Diagonalize A. i.e., find non-singular matrix P such that $P^{-1}AP = D$. $\therefore A = PDP^{-1}$

i) To find A^n :
$$A^n = P \begin{bmatrix} \lambda_1^n & 0 & \cdots & 0 \\ 0 & \lambda_2^n & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^n \end{bmatrix} P^{-1}$$

ii) To find e^A :
$$\begin{aligned} e^A &= I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots \\ &= PIP^{-1} + PDP^{-1} + P \frac{D^2}{2!} P^{-1} + P \frac{D^3}{3!} P^{-1} + \cdots \\ &= P \left(I + D + \frac{D^2}{2!} + \frac{D^3}{3!} + \cdots \right) P^{-1} = P e^D P^{-1} \end{aligned}$$

$$e^A = P \begin{bmatrix} e^{\lambda_1} & 0 & \cdots & 0 \\ 0 & e^{\lambda_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{\lambda_n} \end{bmatrix} P^{-1}$$



Functions of a matrix

iii) To find $\sin A$:

$$\begin{aligned}\sin A &= A - \frac{A^3}{3!} + \frac{A^5}{5!} - \frac{A^7}{7!} + \dots \\ &= P D P^{-1} - P \frac{D^3}{3!} P^{-1} + P \frac{D^5}{5!} P^{-1} - P \frac{D^7}{7!} P^{-1} + \dots \\ &= P \left(D - \frac{D^3}{3!} + \frac{D^5}{5!} - \frac{D^7}{7!} + \dots \right) P^{-1} = P \sin D P^{-1}\end{aligned}$$

$$\sin A = P \begin{bmatrix} \sin \lambda_1 & 0 & \dots & 0 \\ 0 & \sin \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sin \lambda_n \end{bmatrix} P^{-1}$$



Functions of a matrix

iii) To find $\cos A$:

$$\begin{aligned}\cos A &= I - \frac{A^2}{2!} + \frac{A^4}{4!} - \frac{A^6}{6!} + \dots \\ &= P I P^{-1} - P \frac{D^2}{4!} P^{-1} + P \frac{D^4}{4!} P^{-1} - P \frac{D^6}{6!} P^{-1} + \dots \\ &= P \left(I - \frac{D^2}{2!} + \frac{D^4}{4!} - \frac{D^6}{6!} + \dots \right) P^{-1} = P \cos D P^{-1}\end{aligned}$$

$$\cos A = P \begin{bmatrix} \cos \lambda_1 & 0 & \dots & 0 \\ 0 & \cos \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \cos \lambda_n \end{bmatrix} P^{-1}$$



Examples

1) If $A = \begin{bmatrix} 1 & 1 \\ 4 & -2 \end{bmatrix}$, find e^A .

Solution: The Characteristic equation is $|A - \lambda I| = 0$.
$$\lambda^2 + \lambda - 6 = 0$$

Eigenvalues are: $\lambda_1 = -3$ and $\lambda_2 = 2$

Eigenvectors are: $X_1 = \begin{bmatrix} 1 \\ -4 \end{bmatrix}$ and $X_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Let $P = \begin{bmatrix} 1 & 1 \\ -4 & 1 \end{bmatrix}$ and $P^{-1} = \frac{1}{5} \begin{bmatrix} 1 & -1 \\ 4 & 1 \end{bmatrix}$. Then $P^{-1}AP = D = \begin{bmatrix} -3 & 0 \\ 0 & 2 \end{bmatrix}$

$$e^A = P \begin{bmatrix} e^{\lambda_1} & 0 \\ 0 & e^{\lambda_2} \end{bmatrix} P^{-1} = \begin{bmatrix} 1 & 1 \\ -4 & 1 \end{bmatrix} \begin{bmatrix} e^{-3} & 0 \\ 0 & e^2 \end{bmatrix} \cdot \frac{1}{5} \begin{bmatrix} 1 & -1 \\ 4 & 1 \end{bmatrix} = \begin{bmatrix} 5.92 & 1.46 \\ 5.87 & 1.51 \end{bmatrix}$$



Example

2. Find the value of $\cos A$ if $A = \begin{bmatrix} 8 & -4 \\ 2 & 2 \end{bmatrix}$.

Solution: The Characteristic equation is $|A - \lambda I| = 0$.

$$\begin{vmatrix} 8 - \lambda & -4 \\ 2 & 2 - \lambda \end{vmatrix} = 0$$
$$\lambda^2 - 10\lambda + 24 = 0$$

Eigenvalues are: $\lambda_1 = 4$ and $\lambda_2 = 6$

Eigenvectors are: $X_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $X_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

Let $P = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}$ and $P^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}$. Then $P^{-1}AP = D = \begin{bmatrix} 4 & 0 \\ 0 & 6 \end{bmatrix}$

$$\begin{aligned} \cos A &= P \cos D P^{-1} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} \cos 4 & 0 \\ 0 & \cos 6 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} \cos 4 & \cos 6 \\ \cos 4 & 2 \cos 6 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 \cos 4 - \cos 6 & -\cos 4 + \cos 6 \\ 2 \cos 4 - 2 \cos 6 & -\cos 4 + 2 \cos 6 \end{bmatrix} = \begin{bmatrix} -2.2674 & 1.6138 \\ -3.2276 & 2.5739 \end{bmatrix} \end{aligned}$$



Minimal Polynomial

- Let $f(\lambda)$ be a polynomial in λ and A be a square matrix of order n . If $f(A) = 0$, then we say the polynomial $f(\lambda)$ **annihilates (kills)** the matrix A .
- A polynomial in x in which the coefficient of the highest power of x is unity is called a **monic polynomial**.
Ex. $x^3 - 3x^2 - 4x + 4$ is a monic polynomial of degree 3.
- **The characteristic polynomial $|A - \lambda I|$ always annihilates matrix A . (Cayley Hamilton Theorem)**

Definition: The non-zero monic polynomial of lowest degree which annihilates a matrix is called the **minimal polynomial** of that matrix.

Minimal polynomial denoted by $m(\lambda)$.

Characteristic polynomial $|A - \lambda I|$ is denoted by $f(\lambda)$.



Properties of Minimal Polynomial

- i. $\deg m(\lambda) \leq \deg f(\lambda)$
- ii. The minimal polynomial $m(\lambda)$ of A is divisor of every polynomial $P(\lambda)$ where $P(A) = 0$.
- iii. Some or all eigenvalues of A satisfy the minimal polynomial equation $m(\lambda) = 0$.
- iv. If minimal polynomial $m(\lambda)$ is identical with the characteristic polynomial $f(\lambda)$, the matrix is said to be **non-derogatory**.
- v. If $\deg m(\lambda) < \deg f(\lambda)$, then the matrix is said to be **derogatory**.



Minimal Polynomial

Method to find minimal polynomial:

1. Consider first degree monic polynomial $m_1(\lambda) = (\lambda - a)$
If $m_1(\lambda)$ is the minimal polynomial of A , then $m_1(A) = 0$
i.e., $A - aI = 0 \Rightarrow A = aI$
If $m_1(A) \neq 0$, then $m_1(\lambda)$ is not the minimal polynomial of A and we proceed to next higher degree.
2. Consider second degree monic polynomial $m_2(\lambda) = \lambda^2 - a\lambda - b$
If a and b can be uniquely determined such that $m_2(A) = A^2 - aA - bI = 0$,
then $m_2(\lambda)$ is a minimal polynomial otherwise proceed to next higher degree.
3. Consider third degree monic polynomial $m_3(\lambda) = \lambda^3 - a\lambda^2 - b\lambda - c$
If a, b, c can be uniquely determined such that $m_3(A) = A^3 - aA^2 - bA - cI = 0$,
then $m_3(\lambda)$ is a minimal polynomial, else proceed to next highest degree.
4. Continue the procedure up to $(n - 1)$ degree polynomial. If $(n - 1)$ th degree polynomial is not minimal polynomial, then the Characteristic polynomial is the Minimal Polynomial.



Examples

1) Find the minimal polynomial of the matrix $A = \begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$.

Solution:

Consider first degree monic polynomial $m_1(\lambda) = (\lambda - a)$

$$m_1(A) = (A - aI) \neq 0 \quad (\text{since } A \neq aI)$$

Therefore $m_1(\lambda)$ is not a minimal polynomial.

Consider second degree monic polynomial $m_2(\lambda) = \lambda^2 - a\lambda - b$

$$m_2(A) = 0 \Rightarrow A^2 - aA - bI = 0 \Rightarrow A^2 = aA + bI$$

$$\begin{bmatrix} 5 & 3 & 1 \\ 4 & 2 & 3 \\ 4 & -2 & 7 \end{bmatrix} = a \begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix} + b \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2a + b & -2a & 3a \\ a & a + b & a \\ a & 3a & -a + b \end{bmatrix}$$



Examples

$$\begin{aligned} 2a + b &= 5; & a &= 4; & a &= 4; \\ -2a &= 3; & a + b &= 2; & 3a &= -2; \\ 3a &= 1; & a &= 3; & -a + b &= 7; \end{aligned}$$

Since a & b cannot be uniquely determined from above equations, $m_2(\lambda)$ is not a minimal polynomial.

Therefore, Characteristic polynomial of A is the minimal polynomial of A .

$$\text{Now characteristic polynomial of } A = |A - \lambda I| = \begin{vmatrix} 2 - \lambda & -2 & 3 \\ 1 & 1 - \lambda & 1 \\ 1 & 3 & -1 - \lambda \end{vmatrix}$$

$$= \lambda^3 - 2\lambda^2 - 5\lambda + 6$$

$\therefore \lambda^3 - 2\lambda^2 - 5\lambda + 6$ is the minimal polynomial of A .

Therefore the matrix A is non-derogatory.



Examples

2) Find the minimal polynomial of the matrix $A = \begin{bmatrix} 7 & 4 & -1 \\ 4 & 7 & -1 \\ -4 & -4 & 4 \end{bmatrix}$.

Solution:

Consider first degree monic polynomial $m_1(\lambda) = (\lambda - a)$

$$m_1(A) = (A - aI) \neq 0 \quad (\text{since } A \neq aI)$$

Therefore $m_1(\lambda)$ is not a minimal polynomial.

Consider second degree monic polynomial $m_2(\lambda) = \lambda^2 - a\lambda - b$

$$m_2(A) = 0 \Rightarrow A^2 - aA - bI = 0 \Rightarrow A^2 = aA + bI$$

$$\begin{bmatrix} 69 & 60 & -15 \\ 60 & 69 & -15 \\ -60 & -60 & 24 \end{bmatrix} = a \begin{bmatrix} 7 & 4 & -1 \\ 4 & 7 & -1 \\ -4 & -4 & 4 \end{bmatrix} + b \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 7a + b & 4a & -a \\ 4a & 7a + b & -a \\ -4a & -4a & 4a + b \end{bmatrix}$$



Examples

$$7a + b = 69; \quad 4a = 60; \quad -a = -15;$$

$$4a = 60; \quad 7a + b = 69; \quad -a = -15;$$

$$-4a = -60; \quad -4a = -60; \quad 4a + b = 24;$$

$a = 15$ and $b = -36$ satisfy all the equations.

$\therefore m_2(\lambda) = \lambda^2 - 15\lambda + 36$ is the minimal polynomial of A.

Since $\deg m(\lambda) < \deg f(\lambda)$, therefore the matrix A is derogatory.

THANK YOU!